

KOEbenhavns AP, Denmark

WMO: 061800

Lat: 55.614N

Lon: 12.645E

Elev: 17

StdP: 14.69

Time Zone: 1.00 (EUC)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	19.6	22.6	12.4	10.4	24.1	15.7	12.2	25.6	33.8	41.3	31.0	39.5	10.1	50	0.670

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	13.7	78.4	65.3	75.5	64.4	73.0	63.2	67.8	74.3	66.1	72.4	64.7	70.9	10.2	160

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
65.4	93.9	71.1	63.8	88.8	69.2	62.3	84.0	67.8	32.1	74.2	30.8	72.2	29.7	70.9	74.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 27.5	24.7	22.3	DB	13.8	83.2	5.0	2.7	10.2	85.1	7.3	86.7	4.5	88.3	0.9	90.3	(4)
(5)			WB	13.1	70.5	4.8	2.0	9.7	72.0	6.9	73.1	4.2	74.3	0.7	75.7	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	48.3	33.9	34.2	37.8	45.3	53.1	59.3	64.3	64.0	57.7	49.6	42.3	36.7		
	DBStd	12.17	6.05	5.95	5.68	5.58	5.42	4.69	4.91	4.48	4.28	5.20	5.41	6.40		
	HDD50	2229	498	443	380	161	28	0	0	0	2	70	235	412		
	HDD65	6216	963	863	845	591	369	180	75	75	220	476	682	877		
	CDD50	1596	0	0	0	20	125	280	442	434	233	58	4	0		
	CDD65	108	0	0	0	0	1	10	52	43	2	0	0	0		
	CDH74	410	0	0	0	0	5	40	230	134	1	0	0	0		
	CDH80	34	0	0	0	0	0	4	24	6	0	0	0	0		
	WSAvg	11.8	13.5	13.4	12.5	10.9	10.5	10.8	9.9	10.3	11.4	12.5	12.7	13.1		
	PrecAvg	24.5	1.9	1.3	1.3	1.2	1.9	2.5	2.4	3.1	2.3	2.6	2.1	2.1		
(15) Precipitation	PrecMax	32.9	3.4	3.0	2.5	2.9	3.3	5.4	6.9	7.2	6.1	5.1	5.7	3.8		
	PrecMin	16.3	0.0	0.2	0.3	0.3	0.4	0.1	0.1	0.7	0.5	1.0	0.2	0.8		
	PrecStd	3.9	0.9	0.6	0.7	0.6	0.7	1.1	1.6	1.9	1.5	1.1	1.1	0.8		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	48.4	49.6	55.9	66.4	73.7	78.1	82.7	80.6	73.1	62.6	55.7	50.7		
		MCWB	46.1	45.7	48.1	55.5	61.2	64.9	66.9	65.7	63.9	57.9	53.7	49.2		
	2%	DB	45.7	46.5	51.2	61.3	69.6	74.0	79.4	77.4	69.4	60.1	53.3	48.6		
		MCWB	43.9	44.4	45.0	51.8	58.6	62.6	65.6	65.4	61.6	56.7	51.5	47.0		
	5%	DB	43.8	44.4	48.1	57.5	66.1	70.7	76.5	75.0	66.6	58.3	51.4	46.7		
		MCWB	42.2	42.3	43.7	49.7	56.5	60.8	65.0	64.5	60.0	55.2	49.6	45.2		
	10%	DB	42.1	42.4	45.8	54.4	62.8	67.9	73.5	72.0	64.6	56.7	49.7	44.9		
		MCWB	40.7	40.5	42.4	48.0	54.8	59.2	63.6	63.3	58.9	53.9	47.8	43.5		
	0.4%	WB	46.7	46.9	49.4	57.4	63.4	67.5	70.2	69.8	65.4	59.5	54.4	49.6		
		MCDB	47.8	48.2	54.5	64.3	71.7	74.7	78.1	76.2	70.6	61.2	55.5	50.3		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	44.2	44.6	46.3	53.2	60.2	64.4	68.2	67.6	62.9	57.3	51.6	47.4		
		MCDB	45.2	45.9	49.2	59.5	66.8	71.2	75.2	73.4	67.3	59.5	52.9	48.4		
	5%	WB	42.4	42.4	44.4	50.9	57.8	62.4	66.5	65.9	61.4	55.8	49.8	45.4		
		MCDB	43.4	43.7	47.1	56.1	64.0	68.8	73.4	71.6	65.2	57.7	51.0	46.4		
	10%	WB	40.6	40.5	42.7	48.8	55.7	60.5	65.0	64.6	60.0	54.4	48.1	43.6		
		MCDB	41.7	41.9	45.2	53.3	61.2	66.1	71.8	70.2	63.5	56.3	49.3	44.7		
	5% DB	MDBR	6.6	7.3	10.3	13.3	14.0	13.4	13.7	12.9	10.9	8.6	6.7	6.5		
		MCDBR	7.2	8.9	13.9	17.8	18.7	17.0	17.4	15.9	12.9	9.4	6.9	7.0		
		MCWBR	6.9	7.0	9.3	10.5	10.2	8.8	7.6	7.1	7.3	7.1	6.3	6.9		
		MCWBR	7.2	8.6	13.0	15.9	16.1	15.4	15.3	13.9	11.7	8.7	6.6	6.9		
(40) Clear-Sky Solar Irradiance	taub		0.315	0.337	0.360	0.393	0.389	0.383	0.405	0.405	0.374	0.357	0.339	0.306		
		taud	2.287	2.306	2.300	2.275	2.343	2.384	2.343	2.351	2.419	2.398	2.331	2.260		
	Ebn at Noon		188	230	254	262	271	273	264	256	248	221	177	161		
		Edh at Noon	17	24	31	37	36	36	36	34	28	23	17	14		
(44) All-Sky Solar Radiation	RadAvg		142	334	771	1325	1707	1815	1686	1356	960	471	180	95		
	RadStd		27	53	86	162	170	141	205	100	94	69	32	9		

Historical Trends

		Heating		Cooling		Degree-Days					
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65	
(46)	Station Only	(d) +1.30	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) -295	(k) -434	(l) +157	(m) N/A
(47)	Regional (62 neighbors)	+0.74	N/A	N/A	N/A	N/A	N/A	-304	+112	N/A	

Nomenclature: See separate page